

FarmConnect Class Diagram (PlantUML Code)

```
@startuml
skinparam classAttributeIconSize 0
```

```
enum Role {
    PRODUCER
    BUYER
    ADMIN
}
```

```
enum UserStatus {
    ACTIVE
    SUSPENDED
    PENDING_VERIFICATION
    DELETED
}
```

```
enum VerificationStatus {
    UNVERIFIED
    PENDING
    APPROVED
    REJECTED
}
```

```
enum ProductStatus {
    DRAFT
    ACTIVE
    INACTIVE
    OUT_OF_STOCK
    ARCHIVED
}
```

```
enum OrderStatus {
    PENDING
    CONFIRMED
    PROCESSING
    SHIPPED
    DELIVERED
    CANCELLED
    REFUNDED
}
```

```
enum PaymentStatus {
    UNPAID
    PENDING
    PAID
    FAILED
    REFUNDED
    PARTIALLY_REFUNDED
}
```

```
enum DeliveryStatus {
    NOT_SHIPPED
    PREPARING
    IN_TRANSIT
    OUT_FOR_DELIVERY
    DELIVERED
    FAILED_DELIVERY
    RETURNED
}
```

```
enum DeliveryMethod {
    PICKUP
    DELIVERY
}
```

```
enum BuyerBusinessType {
    INDIVIDUAL
    RESTAURANT
    HOTEL
    WHOLESALE
    RETAIL
    OTHER
}
```

```
enum PaymentMethod {
```

FarmConnect Class Diagram (PlantUML Code)

```
CASH_ON_DELIVERY
BANK_TRANSFER
CIB_CARD
EDAHABIA
}

enum PaymentEventType {
    INITIATED
    AUTHORIZED
    CAPTURED
    FAILED
    REFUNDED
    CHARGEBACK
}

enum PayoutStatus {
    PENDING
    PROCESSING
    PAID
    FAILED
}

enum NotificationType {
    ORDER_PLACED
    ORDER_CONFIRMED
    ORDER_SHIPPED
    ORDER_DELIVERED
    ORDER_CANCELLED
    PAYMENT_SUCCESS
    PAYMENT_FAILED
    PRODUCT_LOW_STOCK
    PRODUCER_VERIFIED
    PRODUCER_REJECTED
    GENERAL
}

enum NotificationChannel {
    IN_APP
    EMAIL
    SMS
    PUSH
}

enum AuditAction {
    CREATE
    UPDATE
    DELETE
    LOGIN
    LOGOUT
    VERIFY
    REJECT
    SUSPEND
    UNSUSPEND
}

enum CouponType {
    PERCENT
    FIXED
}

enum ReportTargetType {
    USER
    PRODUCT
    ORDER
}

enum ReportReason {
    SPAM
    FRAUD
    ABUSE
    INAPPROPRIATE_CONTENT
    OTHER
}

enum ReportStatus {
```

FarmConnect Class Diagram (PlantUML Code)

```
    OPEN
    REVIEWING
    RESOLVED
    REJECTED
}

enum ProductUnit {
    KG
    PIECE
    BOX
}

enum ProductLogType {
    WATERING
    HARVEST
    FERTILIZE
    OTHER
}

class User {
    id: String
    email: String
    passwordHash: String
    fullName: String
    role: Role
    status: UserStatus
    avatarUrl: String?
    phone: String?
    createdAt: DateTime
    updatedAt: DateTime
    deletedAt: DateTime?
    refreshTokens: RefreshToken[]
    producerProfile: ProducerProfile?
    buyerProfile: BuyerProfile?
    addresses: Address[]
    cart: Cart?
    orders: Order[]
    reviews: Review[]
    favoriteProducts: FavoriteProduct[]
    favoriteProducers: FavoriteProducer[]
    reportsSubmitted: Report[]
    verificationReviews: ProducerVerificationRequest[]
    notifications: Notification[]
    aiRecommendations: AiRecommendation[]
    auditLogs: AuditLog[]
}

class RefreshToken {
    id: String
    token: String
    userId: String
    expiresAt: DateTime
    revokedAt: DateTime?
    userAgent: String?
    ipAddress: String?
    createdAt: DateTime
    user: User
}

class ProducerProfile {
    id: String
    userId: String
    businessName: String
    businessType: String?
    bio: String?
    latitude: Decimal?
    longitude: Decimal?
    producerOffersDelivery: Boolean
    wilaya: String
    commune: String
    nif: String?
    nis: String?
    nifDocumentUrl: String?
    verificationStatus: VerificationStatus
    verifiedAt: DateTime?
```

FarmConnect Class Diagram (PlantUML Code)

```
verifiedById: String?
createdAt: DateTime
updatedAt: DateTime
user: User
products: Product[]
verificationRequests: ProducerVerificationRequest[]
payouts: Payout[]
reviewsReceived: Review[]
favoritedBy: FavoriteProducer[]
coupons: Coupon[]
}

class ProducerVerificationRequest {
    id: String
    producerProfileId: String
    status: VerificationStatus
    notes: String?
    documents: Json
    submittedAt: DateTime
    reviewedAt: DateTime?
    reviewedById: String?
    producerProfile: ProducerProfile
    reviewer: User?
}

class BuyerProfile {
    id: String
    userId: String
    businessType: BuyerBusinessType?
    createdAt: DateTime
    updatedAt: DateTime
    user: User
}

class Address {
    id: String
    userId: String
    label: String
    recipientName: String
    phone: String
    wilaya: String
    commune: String
    street: String
    postalCode: String?
    isDefault: Boolean
    createdAt: DateTime
    updatedAt: DateTime
    user: User
    orders: Order[]
}

class Category {
    id: String
    nameEn: String
    nameAr: String
    slug: String
    parentId: String?
    createdAt: DateTime
    updatedAt: DateTime
    parent: Category?
    children: Category[]
    products: Product[]
    aiForecasts: AiForecast[]
}

class Product {
    id: String
    producerId: String
    categoryId: String?
}

class ProductLog {
    id: String
    productId: String
    type: ProductLogType
```

FarmConnect Class Diagram (PlantUML Code)

```
    note: String
    happenedAt: DateTime
    createdBy: String
    createdAt: DateTime
    product: Product
}

class ProductImage {
    id: String
    productId: String
    url: String
    altText: String?
    position: Int
    createdAt: DateTime
    product: Product
}

class Cart {
    id: String
    buyerId: String
    createdAt: DateTime
    updatedAt: DateTime
    buyer: User
    items: CartItem[]
}

class CartItem {
    id: String
    cartId: String
    productId: String
    quantity: Int
    createdAt: DateTime
    updatedAt: DateTime
    cart: Cart
    product: Product
}

class Order {
    id: String
    buyerId: String
    buyerAddressId: String?
    status: OrderStatus
    paymentStatus: PaymentStatus
    deliveryStatus: DeliveryStatus
    deliveryMethod: DeliveryMethod
    deliveryFee: Decimal?
    deliveryVerificationToken: String?
    verifiedAt: DateTime?
    couponCode: String?
    discountAmount: Decimal?
    total: Decimal
    currency: String
    notes: String?
    createdAt: DateTime
    updatedAt: DateTime
    buyer: User
    buyerAddress: Address?
    items: OrderItem[]
    payment: Payment?
    deliveryTrackings: DeliveryTracking[]
    reviews: Review[]
}

class OrderItem {
    id: String
    orderId: String
    productId: String?
}

class Review {
    id: String
    orderId: String
    productId: String
    producerId: String
    buyerId: String
```

FarmConnect Class Diagram (PlantUML Code)

```
    rating: Int
    comment: String?
    createdAt: DateTime
    order: Order
    product: Product
    producer: ProducerProfile
    buyer: User
}

class FavoriteProduct {
    id: String
    buyerId: String
    productId: String
    createdAt: DateTime
    buyer: User
    product: Product
}

class FavoriteProducer {
    id: String
    buyerId: String
    producerId: String
    createdAt: DateTime
    buyer: User
    producer: ProducerProfile
}

class Coupon {
    id: String
    code: String
    producerId: String
    type: CouponType
    amount: Decimal
    startsAt: DateTime
    endsAt: DateTime
    usageLimit: Int?
    usedCount: Int
    isActive: Boolean
    createdAt: DateTime
    updatedAt: DateTime
    producer: ProducerProfile
}

class Report {
    id: String
    reporterId: String
    targetType: ReportTargetType
    targetId: String
    reason: ReportReason
    description: String
    status: ReportStatus
    internalNote: String?
    createdAt: DateTime
    updatedAt: DateTime
    reporter: User
}

class Payment {
    id: String
    orderId: String
    method: PaymentMethod
    status: PaymentStatus
    amount: Decimal
    currency: String
    gatewayRef: String?
    gatewayResponse: Json?
    paidAt: DateTime?
    failedAt: DateTime?
    createdAt: DateTime
    updatedAt: DateTime
    order: Order
    events: PaymentEvent[]
}

class PaymentEvent {
```

FarmConnect Class Diagram (PlantUML Code)

```
    id: String
    paymentId: String
    type: PaymentEventType
    payload: Json
}

class Payout {
    id: String
    producerProfileId: String
    amount: Decimal
    currency: String
    status: PayoutStatus
    bankName: String?
    bankAccount: String?
    reference: String?
    notes: String?
    processedAt: DateTime?
    createdAt: DateTime
    updatedAt: DateTime
    producer: ProducerProfile
}

class DeliveryTracking {
    id: String
    orderId: String
    status: DeliveryStatus
    location: String?
    description: String?
    occurredAt: DateTime
    order: Order
}

class Notification {
    id: String
    userId: String
    type: NotificationType
    channel: NotificationChannel
    title: String
    body: String
}

class AnalyticsDaily {
    id: String
    date: DateTime
    wilaya: String?
    newUsers: Int
    newProducers: Int
    newBuyers: Int
    totalOrders: Int
    totalRevenue: Decimal
    createdAt: DateTime
    updatedAt: DateTime
}

class AiRecommendation {
    id: String
    userId: String
    productId: String
    score: Float
    reason: String?
    seenAt: DateTime?
    clickedAt: DateTime?
    createdAt: DateTime
    user: User
    product: Product
}

class AiForecast {
    id: String
    productId: String?
    categoryId: String?
    forecastDate: DateTime
    predictedDemand: Decimal
    confidenceScore: Float
    modelVersion: String
}
```

FarmConnect Class Diagram (PlantUML Code)

```
    createdAt: DateTime
    product: Product?
    category: Category?
}

class AuditLog {
    id: String
    actorId: String?
    targetType: String
    targetId: String
    action: AuditAction
}

Address --> User : user
AiForecast --> Category : category
AiForecast --> Product : product
AiRecommendation --> Product : product
AiRecommendation --> User : user
BuyerProfile --> User : user
Cart --> User : buyer
CartItem --> Cart : cart
CartItem --> Product : product
Category --> Category : parent
Coupon --> ProducerProfile : producer
DeliveryTracking --> Order : order
FavoriteProducer --> ProducerProfile : producer
FavoriteProducer --> User : buyer
FavoriteProduct --> Product : product
FavoriteProduct --> User : buyer
Order --> Address : buyerAddress
Order --> Payment : payment
Order --> User : buyer
Payment --> Order : order
Payout --> ProducerProfile : producer
ProducerProfile --> User : user
ProducerVerificationRequest --> ProducerProfile : producerProfile
ProducerVerificationRequest --> User : reviewer
ProductImage --> Product : product
ProductLog --> Product : product
RefreshToken --> User : user
Report --> User : reporter
Review --> Order : order
Review --> ProducerProfile : producer
Review --> Product : product
Review --> User : buyer
User --> BuyerProfile : buyerProfile
User --> Cart : cart
User --> ProducerProfile : producerProfile

@enduml
```